

FABIO NEIRA ALZATE

GIS Specialist | Geospatial Infrastructure - Environmental Projects - Remote Sensing

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PROFESSIONAL SUMMARY

GIS Specialist and Environmental Engineer with 5+ years of experience in corporate geospatial infrastructure and environmental projects combining spatial analysis and satellite remote sensing. Proficient in ESRI platforms (ArcGIS Enterprise, ArcGIS Pro, ArcGIS Online), open source stack (QGIS, GeoServer, PostGIS), and remote sensing tools (Google Earth Engine, SNAP, Sentinel-1 SAR). GIS Specialization (Universidad de Manizales, 2025) with thesis on automated SAR-based waterbody detection in coastal zones. Proven experience in Python/ArcPy automation, multiuser geodatabase administration, and Level 2 GIS user support. Cross-functional work on projects for Cornare, Corantioquia, BanCO2, Metro de Medellín, ISA, and ISAGEN; currently dedicated full-time to Geosura, a national geospatial platform.

CORE COMPETENCIES

ArcGIS Enterprise Admin	QGIS / GeoServer / PostGIS	Remote Sensing (SAR / Sentinel-1)
Python & ArcPy Automation	Google Earth Engine (GEE)	Enterprise Geodatabases
Level 2 GIS User Support	Environmental GIS & Risk Management	OGC Interoperability (WMS/WFS)
Incident Diagnosis & Resolution	ArcGIS Server Administration	Training & Knowledge Transfer

PROFESSIONAL EXPERIENCE

GIS Infrastructure & Support Specialist

Jun 2021 – Present

HYG Consultores S.A.S. | Medellín, Colombia

- Cross-functional GIS support across projects for public and private sector clients: Municipio de Envigado, Municipio de Itagui, Cornare, BanCO2, Corantioquia, Metro de Medellín, ISA, and ISAGEN. Currently dedicated full-time to Geosura, a national geospatial platform, administering its federated ArcGIS Enterprise 11.x environment.
- Automated the full service publishing lifecycle using Python Toolboxes (ArcPy), cutting deployment time by ~60% and eliminating manual errors in the process.
- Diagnose and resolve technical incidents in corporate GIS infrastructure, coordinating with IT teams to guarantee service continuity and geospatial data integrity.
- Designed and implemented a corporate Web GIS methodology based on OGC interoperability standards (WMS/WFS/REST), integrating ESRI services with open source platforms.
- Implemented OAuth2 Client Credentials flow for a public-facing geoportal, managing token lifecycle and service-sharing strategies across federated environments.
- Provide Level 2 technical support to GIS users, resolving configuration, service access, symbology, and performance issues across ArcGIS Pro, ArcGIS Online, and QGIS.
- Developed cartographic applications and web dashboards for non-specialist users, reducing the technical team's dependency for recurring operational queries.
- Led internal GIS training sessions designing learning plans adapted to each team's technical level under Scrum methodology.

GIS Instructor — University Level

Aug 2025 – Present

Politécnico Jaime Isaza Cadavid | Medellín, Colombia

- Design and deliver the GIS for Public Administration course, integrating open source tools (QGIS) and cloud platforms (ArcGIS Online, Google Earth Engine) for territorial management decision-making.
- Develop updated pedagogical materials, currently completing a Diploma in University Teaching to certify the design of organizational learning plans and training programs.

GIS Engineering Support Intern

Aug 2019 – Jun 2021

LINTEK S.A.S. | Colombia

- Processed and managed geographic databases for water and sewage infrastructure systems, supporting data loading and validation on the HydroCaz platform.
- Created, reprojected, and converted vector elements in GIS platforms, ensuring geometric and attribute consistency of delivered datasets.
- Handled topographic data and raster layers (DTMs, orthophotographs) for environmental and hydraulic infrastructure projects.
- Trained end users on the HydroCaz platform, ensuring effective tool adoption across technical teams.

EDUCATION

GIS Specialization (Graduate)

2024 – Jun 2025

Universidad de Manizales

- Advanced spatial analysis, Python automation, satellite imagery processing (ArcGIS Pro, QGIS, SNAP, Google Earth Engine). Thesis: SAR technology (Sentinel-1) for automated waterbody detection in coastal zones.

Environmental Engineering (B.Sc.)

Mar 2021

Universidad de Antioquia

- Complementary studies in environmental modelling, geomatics, risk management, water resource planning, and environmental auditing.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Ongoing Training: AI Agents in Geospatial Workflows

2025 - Present

Practical training -- Claude (Anthropic) - Google AI

- Hands-on training in designing and integrating AI agents into GIS solutions: workflow automation, AI-augmented spatial analysis, and LLM-powered GIS tools (Claude, Google Gemini).

Diploma in University Teaching

2025 – Present

In progress

- Certified to design organizational learning plans and structured training programs for technical and non-technical audiences.

Leadership & Team Management

Mar 2026

Universidad Internacional de La Rioja (UNIR)

- Team management, effective communication, decision-making, and leadership in technical and organizational environments. Credential ID: 77202603011143.

Generative AI Diploma

Jun 2025

Eidos Global + Microsoft

- Foundations and practical applications of generative artificial intelligence in professional and geospatial environments.

Advanced Geospatial Data Analytics in Python

Feb 2026

LinkedIn Learning

- Advanced spatial data analysis and visualization using Python for GIS workflows.

ArcGIS Systems Administration Diploma

Oct 2022

ESRI Colombia

- ArcGIS Enterprise configuration, branch versioning, geodatabase replication, multiuser workflows, and geospatial data management in production environments. Credential ID: 881222.

ArcGIS Enterprise: Administration Workflows

Apr 2024

ESRI Colombia

- Advanced enterprise GIS administration workflows, monitoring, and maintenance. ID: 16261.

Deploying & Maintaining a Multiuser Geodatabase

Aug 2022

ESRI Colombia

- Deployment, configuration, and maintenance of enterprise geodatabases in multiuser environments. IDs: 8736, 8750, 8752, 8902.

Spatial Data Science: The New Frontier in Analytics

Oct 2023

Esri -- Signed by Jack Dangermond

- MOOC on advanced spatial statistics, machine learning, and data-driven decision-making using the ArcGIS platform.

Imagery in Action

Oct 2023

Esri -- Signed by Jack Dangermond

- MOOC on satellite imagery analysis, remote sensing workflows, and image processing with the ArcGIS platform.

Data Science in Python & ArcGIS Pro

Sep 2022

Esri Colombia

- Applied data science workflows using Python integrated with ArcGIS Pro for geospatial analysis and advanced visualization.

Geodatabase (GDB) -- ANLA Environmental Standards

Dec 2022

FG Training Colombia SAS

- 40-hour course on the geodatabase model required by Colombia's National Environmental Licensing Authority (ANLA) for environmental licensing projects.

Remote Sensing & GIS for Flood Risk Management

Apr 2018

Universidad de Antioquia -- Faculty of Engineering

- 32-hour course on satellite remote sensing and GIS applied to flood risk analysis and disaster management. Issued by the Dean of Engineering.

English B1 Certification

Dec 2025

Institución Universitaria ESUMER

- Certified English B1 proficiency with focus on academic and professional technical communication.

AWARDS & RECOGNITIONS

- USAID Leadership Scholarship — Open Data Mapping & Leadership. Training conducted in South Africa, 2019.
- YouthMappers Leadership Fellow — YouthMappers. Recognized for leadership in open data mapping and geospatial volunteering, Jan 2019.
- ACRES Foundation Scholarship — Cámara Colombiana de Infraestructura. Full academic merit scholarship, 2017–2021.
- Lead Member — GeoLab Research Group, Universidad de Antioquia. Geospatial university research and open innovation, 2017–2021.

LANGUAGES

Spanish:

Native

English:

Intermediate (B1 — ESUMER Certified)